

Question number	Answer	Additional guidance	
(a)(i)	A calculation showing the following steps: • correct measurement of Bowman's capsule (1) • calculation of magnification and answer given in standard form (1) OR • conversion of 52 mm to nm (52×1000000) = 5.2×10^7 • calculation of magnification (5.2×10^7) $\div 70$ =	52(mm) (+/-1mm) accept 5.2(cm) (+/- 0.1cm) $(52 \div 0.00007) = 7.43 \times 10^5$ IF THEY USE UNITS eg nm NO MP2 ACCEPT RANGE 7.29×10^5 - 7.57×10^5 7.3×10^5 - 7.6×10^5	(2)

Question number	Answer	Additional guidance	
(a)(ii)	A calculation showing the following steps: • calculation of volume (1) • and answer given to 2 significant figures (1)	$(4/3 \times 3.14 \times 35 \times 35 \times 35) = 179503 \text{ (nm}^3\text{)}$ Accept number within range (179500 – 179600) $= 180000 \text{ (nm}^3\text{)} / 1.8 \times 10^5$	(2)

		correct answer gains both marks	
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Question number	Answer	Additional guidance	
6(b)(i)	Choose an item. • C (osmoreceptor) is the correct answer A is not the correct answer as baroreceptors do not respond to changes in water potential of the blood. B is not the correct answer as chemoreceptors do not respond to changes in water potential of the blood. D is not the correct answer as thermoreceptors do not respond to changes in water potential of the blood.		(1)

Question number	Answer	Additional guidance	Mark
(b)(ii)	An answer that includes one of the following points: • {osmoregulation / to maintain osmotic balance of cells} (1) • {to maintain water content of cells / prevent dehydration of cells / prevent {crenated / shrinking} of cells / prevent cell lysis} (1) • allow correct functioning of kidney (1)	ACCEPT to prevent water {entering / leaving} by osmosis ACCEPT keep {plasma / blood} isotonic / maintain water potential of blood ACCEPT maintain osmotic pressure ignore cells destroyed ACCEPT prevent cells being hypertonic or hypotonic ignore ultrafiltration	(1)

Question number	Answer	Additional guidance	Mark
6(c)	A description that includes four of the following points: • (plasma concentration) detected by osmoreceptors (1) • blood {volume / pressure} (increase) would be detected by {baroreceptors / pressure receptors} in {carotid artery / aortic arch} (1) • (resulting in) pituitary gland to release less ADH (1) • causing reduction in aquaporins in kidney tubule / decreases permeability of {DCT / collecting duct} (1) • leading to less water being reabsorbed by {kidney / DCT / collecting duct} / (more) {dilute / less concentrated} urine production (1)	ignore permeability of tubules decreases unqualified (must say a correct named tubule) ACCEPT less water reabsorbed into blood (but ignore body) ACCEPT more water excreted in urine} ignore more urine produced	(4)

Question number	Answer	Additional guidance	Mark
6(d)(i)	Choose an item A (co-transport with sodium ions) is the correct answer B is not the correct answer as glucose is not reabsorbed by endocytosis. C is not the correct answer as glucose is not reabsorbed by exocytosis. D is not the correct answer as glucose is not reabsorbed by osmosis.		(1)

Question number	Answer	Additional guidance	Mark
6(d)(ii)	A calculation showing the following steps: - conversion of 7.5 g to mg (1) - calculation of rate per second (1)	$(7.5 \times 1000) / 7500$ 7.5×10^3 $7500 \div (60 \times 60) = 2.08 / 2.083 / 2.1$ ecf applied for mp2 for division by 3600 (60 x 60) correct answer gains both marks	(2)

